MIL-C-60538(MU)
19 August 1966
SUPERSEDING
FA-PD-MI-2704, Rev. 0
13 January 1966

MILITARY SPECIFICATION

CANISTER, 90MM, M336 METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification is the applicable detail specification for Canister, 90MM, M336 Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military
MIL-A-2550 - Ammunition and Special Weapons, General Specification For
MIL-P-10025 - Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification For

STANDARDS
Federal
Federal Test Method Standard No. 151 - Metal; Test Methods

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-639 - Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1315

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DRAWINGS
Munitions Command
F10535895 - Canister, 90MM, M336 Metal Parts Assembly
F7548604 - Pallet - Pallet for Fixed and Semi-Fixed Projectiles, Assembly, Details, Packing and Marking
IEL-10535895 - Inspection Equipment List for Canister, 90MM, M336 Metal Parts Assembly

PUBLICATIONS
Frankford Arsenal
Operating Instructions for Use of Detroit Testing Machine for Band Seat Inspection

(Copies of Specifications, Standards, Drawings, and Publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts, and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 First article sample.- Requirements for the submission of first article sample by the contractor shall be as specified in the contract.

3.3 Rotating band

3.3.1 Band seat.- Prior to application of rotating band blank, the band seat shall be clean and free from oil, rust, dirt, grease or other foreign matter. Shot or sandblasting shall not be permitted for cleaning the band seat after knurling.

3.3.2 Application.- The rotating band shall be applied after heat treatment of the body, by carefully pressing the band blank into the band seat so as to make contact with the sidewalls. There shall be no gap between the rotating band and sidewalls of the band seat after the band has been machined to final dimension.

3.3.3 Band seating.- The diametrical clearance between the rotating band and the band seat shall not exceed .006 inch.

3.3.4 Mechanical properties.- After application and subsequent removal, the band shall have the following mechanical properties:

Tensile strength - 32,000 pounds per square inch (PSI) minimum
Elongation - 15 percent per inch of gage length

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3.3.5 Band damaged.- The machined rotating band shall be carefully protected from being nicked, burred or otherwise damaged in subsequent handling and shipping.

3.4 Metal defects.- The canister body assembly shall be free from cracks, splits, cold shuts, inclusions, porosity or other similar defects.

3.5 Protective coating.- Protective coating shall be in accordance with the requirements of the applicable drawing and specifications listed thereon.

3.6 Painting.- Painting shall comply with the requirements of the applicable drawings. All paint shall be dry to the touch prior to testing. After salt spray exposure, there will be no evidence of corrosion externally.

3.7 Functional

3.7.1 Security.- There shall be no loss of components nor breakup of the canister assembly in the gun bore or muzzle break, as evidenced by damage to the muzzle brake by component parts of the canister assembly or by components thereof being deflected to the rear of the muzzle brake. The canister shall effectively separate forward of the gun muzzle.

3.8 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.1 Burr.- No part shall have a burr which might interfere with the assembly or function of the round, or which might be injurious to personnel handling the item.

3.8.2 Foreign matter.- No part or assembly shall contain dirt, chips, grease, oil, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define Quality Assurance Terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample

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4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of thirty-two canister assemblies as follows:

20 canister assemblies complete (painted)
3 canister assemblies complete (unpainted)
3 body and band assemblies (unpainted)
3 bodies unpainted
3 sets of component parts except the slugs, unassembled and unpainted including band blanks (bases to be completely machined, including band seat)

Fifteen of the above painted canister assemblies shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions

4.3.1 Lot

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105 or MIL-STD-1235, as applicable. An AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed by MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

4.3.2.1 Disc, closing, Drawing B10535902 covering a detail of Drawing F10535895.

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4.3.2.1 (Cont'd)

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter	Gage
MINOR	
201. Length, total	Gage
202. Thickness through disc	Gage
203. Radii or chamfers missing or incorrect	Visual (see 6.1)
204. Surface finish improper	Visual (see 6.1)
205. Burr	Visual
206. Foreign matter	Visual

4.3.2.2 Tube, Drawing D10535898 and tube, alternative, Drawing D10535904 covering a detail of Drawing F10535895.

CRITICAL	
None defined	
MAJOR	
101. Diameter of rear counterbore	Gage
102. Depth of rear counterbore	Gage
103. Welded seam cracked or otherwise defective (if applicable)	Visual
104. Metal defective	Visual
MINOR	
201. Location of weld slots	Gage
202. Length of weld slot(s)	Gage
203. Width of weld slot(s)	Gage
204. Radii or chamfers missing or incorrect	Visual (see 6.1)
205. Surface finish, improper	Visual (see 6.1)
206. Burr	Visual
207. Foreign matter	Visual

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4.3.2.3 Base, Drawing C10535899 covering a detail of Drawing F10535895

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Length from rear face to front face of tube seat rib (if applicable)	Gage
102. Metal defective	Visual
MINOR	
201. Diameter forward of rib	Gage
202. Depth of drill points max. (if applicable)	Gage
203. Radii or chamfers missing or incorrect	Visual (See 6.1)
204. Surface finish improper	Visual (See 6.1)
205. Burr	Visual
206. Foreign matter	Visual

4.3.2.4 Body, Drawing D10535897 and body alternative, Drawing D10535903 covering a detail of Drawing F10535895.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of rotating band seat with tube	Gage
102. Diameter of rotating band seat	Gage
103. Location of rotating band seat	Gage
104. Width of rotating band seat	Gage
105. Metal defective	Visual
MINOR	
201. Length, total	Gage
202. Scoring missing from rotating band seat	Visual
203. Radii or chamfers missing or incorrect	Visual (See 6.1)
204. Evidence of shot or sandblasting on rotating band seat	Visual
205. Burr	Visual
206. Foreign matter	Visual

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4.3.2.5 Body and band assembly, Drawing D10535896 covering a detail of Drawing F10535895.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Diameter of counterbore at front of tube	Gage
102.	Diameter, inside tube, min.	Gage
103.	Length of bursting grooves	Gage
104.	Locating of bursting slots	Gage
105.	Thickness of tube wall, min.	Gage
106.	Thickness of counterbore wall, min.	Gage
107.	Thickness of wall at bursting grooves	Gage
108.	Thickness through base	Gage
109.	Weld(s) cracked or otherwise defective	Visual
110.	Protective coating inadequate or damaged	Visual
111.	Seam sealer not visible for full 360 degrees at juncture of base and tube wall	Visual
MINOR		
201.	Depth of counterbore at front of tube	Gage
202.	Width of bursting grooves	Gage
203.	Radii or chamfers, missing or incorrect	Visual (See 6.1)
204.	Surface finish improper	Visual (See 6.1)
205.	Burr	Visual
206.	Foreign matter	Visual

4.3.2.6 Slug, Drawing B10535900, covering a detail of Drawing F10535895.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Surface finish improper	Visual (See 6.1)
202.	Protective coating inadequate or damaged	Visual
203.	Burr	Visual
204.	Foreign matter	Visual

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4.3.2.7 Canister assembly, Drawing F10535895 (before paint)

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Concentricity of rotating band with tube	Gage
102. Concentricity of bourrelet with tube	Gage
103. Concentricity of base cavity with tube	Gage
104. Straightness of tube	Gage
105. Diameter of rotating band	Gage
106. Diameter of tube, max.	Gage
107. Diameter of bourrelet, min.	Gage
108. Depth of crimp groove	Gage
109. Location of rotating band	Gage
110. Location of crimp groove	Gage
111. Location of angle on rotating band	Gage
112. Length, total, max.	Gage
113. Profile of radius rear of crimp groove	Gage
114. Radius of crimp groove	Gage
115. Gap between base rib and tube, max. (if applicable)	Gage
116. Closing disc improperly soldered	Visual
MINOR	
201. Diameter of rotating band relief groove	Gage
202. Diameter of base cavity	Gage
203. Width of rotating band	Gage
204. Width of rotating band relief groove	Gage
205. Depth of undercut front and rear of rotating band (if applicable)	Visual (See 6.1)
206. Width of undercut front and rear of rotating band (if applicable)	Visual (See 6.1)
207. Radii or chamfers missing or incorrect	Visual (See 6.1)
208. Surface finish improper	Visual (See 6.1)
209. Burr	Visual
210. Foreign matter	Visual

4.3.2.8 Canister assembly, Drawing F10535895 (after paint).

CRITICAL

None defined

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Weight	Approved scale
102. Diameter of bourrelet, max.	Gage
103. Rotating band damaged	Visual
MINOR	
201. Paint on rotating band bearing surfaces	Visual
202. Exterior paint inadequate or damaged	Visual
203. Paint not dry	Visual & Tactile
204. Marking missing, incorrect or illegible	Visual

4.3.2.9 Pallet, Drawing F7548604

CRITICAL

None defined

MAJOR

None defined

MINOR	
201. Pallet damaged (unserviceable)	Visual
202. Sleeve missing	Visual
203. Improperly packed	Visual
204. Strapping missing, loose or broken	Visual & Manual
205. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with 4.4.

4.3.3.1 Mechanical properties.

4.3.3.1.1 Tension test.

4.3.3.1.1.1 Tube.- Two tubes from each tubing lot or fraction thereof shall be subjected to this test to determine compliance with the mechanical properties requirements of the applicable drawing. Failure of one or more tension test specimens to comply with yield strength or elongation requirements shall be cause for rejection of the lot of tubing.

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4.3.3.1.1.2 Rotating band.- One body and hand assembly shall be selected from each 8 hours production or fraction thereof per banding machine in operation and subjected to this test to determine compliance with the requirements of 3.3.4. Failure to comply shall be cause for rejection of the production represented by the sample. When the destructive method for band seating inspection is employed, the band that has been removed for the band seating test may be subjected to this test.

4.3.3.2 Rotating bands.

4.3.3.2.1 Band seating (Destructive method).- One body and band assembly shall be selected from each 8 hour production or fraction thereof for each banding machine in operation (but in nc case shall the quantity selected for this test be less than 2 body and band assemblies per shift). Failure of the band to comply with 3.3.3 shall be cause for rejection of the production represented by the sample.

4.3.3.2.2 Band seating (Nondestructive method).- When the nondestructive method is employed, two body and band assemblies from each one hour's production per banding machine in operation shall be subjected to this test. Failure of any band to comply with 3.3.3 shall be cause for rejection of the lot.

4.3.3.2.3 Sidewall contact.- Rotating bands removed as specified in 4.3.3.1.1.2 or 4.3.3.2.1 shall be examined to determine compliance with 3.3.2. Failure to comply shall be cause for rejection of the assemblies represented by the sample.

4.3.3.3 Strength of welds.

4.3.3.3.1 Tube.- Two tubes from each 8 hour production or fraction thereof per welding machine in operation shall be subjected to this test (if applicable) to determine compliance with the requirements of the applicable drawing. Failure of either tube to comply shall be cause for rejection of the production represented by the failing tube.

4.3.3.3.2 Tube to base.- Two body and band assemblies from each 8 hour production or fraction thereof per welding machine in operation shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure of either body and band assembly to comply shall be cause for rejection of the production represented by the sample. The body and band assemblies used to perform the band seating test may also be used for this test.

4.3.3.4 Cleaning and application of protective coating.- The sample size and acceptance criteria for the inspection of cleaning the item and the application of protective coating shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action recommended therein.

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4.3.4 Functional.- The following proving ground tests will be performed by the Government.

4.3.4.1 First article sample.

4.3.4.1.1 Initial production sample.- The quantity listed in Table I shall be ballistically tested in the phase indicated.

TABLE I

Phase	Quantity
Security	15

4.3.4.2 Production lot sample.- The quantity listed in Table II shall be ballistically tested in the phase indicated.

TABLE II

Phase	Quantity
Security	5

Failure of any canister assembly to comply with the requirements of 3.7.1 shall be cause for rejection of the lot.

4.3.5. Inspection equipment.- Index of Inspection Equipment List IEL-10535895 identified the inspection equipment required to perform the examination and tests prescribed herein. The equipment design shall be in accordance with code designation on the applicable IkEL.

4.4 Test methods and procedures.

4.4.1 Mechanical properties.

4.4.1.1 Tension test.

4.4.1.1.1 Tube.- From the samples provided by 4.3.3.1.1.1 two tension test specimens conforming to the largest obtainable "T" type specimen as prescribed by Federal Test Method Standard No. 151, shall be taken longitudinally from each side of the tube, 90 degrees from the welded seam. The tension test shall be performed using the equipment and criteria for performing the test as prescribed by Federal Test Method Standard No. 151. Yield strength shall be determined by the offset method. The limiting set shall be 0.1 percent (0.001 inch per inch of gage length).

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4.4.1.1.2 Rotating band.- The sample provided by 4.3.3.1.1.2 shall have the rotating band removed and carefully flattened without hammering so as to hold additional cold working to a minimum. A test specimen conforming to the largest obtainable "F" type specimen as prescribed by Federal Test Method Standard No. 151, shall be removed and tested in accordance with the procedures for performing this test as prescribed by Federal Test Method Standard No. 151.

4.4.2 Band seating.

4.4.2.1 Destructive method.- The body and band assembly selected in accordance with 4.3.3.1.1.2 should not be processed through the band machining line, although each band should have a light surface cut to true up the surface prior to measuring. The diameter of each band shall be measured on 3 locations, 60 degrees apart, on the points along the cylindrical portion of their length, near the front and rear edges. This comprises 6 diametrical measurements per band. These points of measurement shall be marked on the band end of the body adjacent thereto in such manner as to permit proper body and band orientation after removal. The bands shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat the sum of the corresponding 2 thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band. The 6 results shall be totaled and the sum thus obtained will be divided by 6 to determine the average diametrical clearance.

4.4.2.2 Nondestructive method.- The test shall be performed on a Detroit Band Testing Machine. The indentor shall be as described on the inspection equipment list. The test procedure shall be in accordance with operating instructions for use of Detroit Testing Machine for band seating inspection (See 6.3).

4.4.2.3 Sidewall contact.- Samples supplied in accordance with 4.3.3.2.3 shall be visually examined for contact with the sidewalls of the band seat.

4.4.3 Strength of weld.

4.4.3.1 Tube.- The samples supplied in accordance with 4.3.3.3.1 (if applicable) shall be flattened between parallel plates to less than 2/3 of its original outside diameter and the weld zone examined for cracks other than superficial surface ruptures. Then the application of pressure will be continued until the tube has been flattened to less than 1/3 of its original outside diameter after which the tube will be examined for cracks, in the heat affected zone.

4.4.3.2 Tube to base.- The samples supplied in accordance with 4.3.3.3.2 shall be mounted in an approved device in such a manner that the tube portion of the body and band assembly will be held stationary while the force is applied to the base. After the force has been released, the body and band assembly shall be visually examined for evidence of separation, cracking of welds or deformation in the heat affected zone.

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4.4.4 Protective coating.- The procedures for testing the cleaning and surface treatment methods for the components and assemblies shall be in accordance with the methods and criteria for testing as specified in the applicable specifications. The salt spray prescribed shall be performed in compliance with Federal Test Method Standard No. 151. A 20 percent sodium chloride solution shall be used.

4.4.5 Functional.

4.4.5.1 Security.- Initial production sample.- The canister assemblies supplied in accordance with 4.3.4.1.1 will be assembled into complete rounds and fired in a gun for which the round is standard. The canister assemblies will be temperature conditioned for 24 hours as follows:

5 rounds will be temperature conditioned at 130 degrees F. to 140 degrees F.

5 rounds will be temperature conditioned at minus 60 degrees F. to minus 65 degrees F.

5 rounds will be fired at ambient temperature

Camera, target and plywood panels will be positioned as directed by the proof officer, functioning, dispersion and range characteristics will be recorded for each round fired for informational purposes.

4.4.5.2 Production lot.- The canister assemblies supplied in accordance with 4.3.4.2 Table II will be assembled into complete rounds and fired in a gun for which the round is standard. The weight of the propellant charge will be that used in service firing. The rounds will be fired at ambient temperature. A smear camera to record the flight of the canister will be used to determine compliance with 3.7.1.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 Level A - Not applicable

5.1.2 Level B - Not applicable

5.1.3 Level C - Packing and marking shall be in accordance with Drawing F7548604 and MIL-P-10025.

5.2 Data card.- Data card information shall be in accordance with MIL-STD-1167.

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6. NOTES

6.1 Visual examination qualification.- When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Band seating.- The operating procedures for use of Detroit Testing Machine, as prepared by the Inspection Equipment Branch, Frankford Arsenal, SMUFA-Q3300, may be used as an aid in checking band seating inspection procedures.

6.3 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project no. 1315-A505

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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